

Abstract Algebra

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"Gates"

Abstraction

- think about a general operation on a set
- prove things that apply in these settings & many more.

Algebra you've seen:

Addition & multiplication on:

$\dots, -2, -1, 0, 1, 2, \dots$ $-\frac{1}{\sqrt{2}}, \pi, e, \sqrt{2}$ $\{a+bi \mid a, b \in \mathbb{R}\}$

• $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{R}^+, \mathbb{C}$

$\{\frac{a}{b} \mid a, b \in \mathbb{Z}, b \neq 0\}$ $\{r \in \mathbb{R} \mid r > 0\}$

- mod n
- on functions $\cos x + \sin x$
- on $n \times n$ matrices $AB, A+B$

Q: How are these operations similar & different?

Binary Operations

Section 2

Def: Let S be a set. A (binary) operation $*$ is a function $*: S \times S \rightarrow S$ written $*(a,b) = a * b$

two inputs

Warning: $*$ must be defined $\forall a,b \in S$, and result must land back in S .

Examples:

- everything from the last slide
- composition of functions $S = \{f: A \rightarrow A\}$, $f \circ g = g \circ f$.

Non-Examples:

- $+$ on all matrices
- $-$ on $\mathbb{R}^+$ ($+$, $\times$, or $\div$ okay!)
- $\times$ on $n \times m$ matrices, $n \neq m$
- $\div$ on $\mathbb{Z}$

Def: A subset $H \subseteq S$ is **closed under** a binary operation $*$ on S if:
 $\forall h_1, h_2 \in H$ we have $h_1 * h_2 \in H$.

In this case we can view $*$ as giving an **induced** binary operation on H .

Example: Binary operation $+$ on $\mathbb{R}$, $\mathbb{Z} \subset \mathbb{R}$ or $\mathbb{Q} \subset \mathbb{R}$
are closed under $+$. {nonzero reals}

Non-Example: Binary operation $+$ on $\mathbb{R}$, subset $\mathbb{R}^* \subset \mathbb{R}$
 $2 + (-2) = 0 \notin \mathbb{R}^*$ so $\mathbb{R}^*$ not closed under $+$.

We can now think about abstract properties that a given (binary) operation might have.

Def: An operation $*$ on S is **commutative** if $\forall a, b \in S$ we have

$$a * b = b * a$$

Def: An operation $*$ on S is **associative** if $\forall a, b, c \in S$ we have

$$(a * b) * c = a * (b * c)$$

Examples:

• $-$ on $\mathbb{Z}$: not commutative: $1 - 2 \neq 2 - 1$, not associative

• $+$ on $\text{Mat}_n(\mathbb{R})$: both

 **nxn matrices with real entries**

$$1 - (1 - 1) \neq (1 - 1) - 1$$

$\begin{matrix} \text{"} & \text{"} \\ 1 & -1 \end{matrix}$

Examples:

- $+$ or $\times$ on $\mathbb{Z}$ OR $\mathbb{Q}$ OR $\mathbb{R}$ OR $\mathbb{C}$: both
- $\times$ on $\text{Mat}_n(\mathbb{R})$:
 - not commutative for $n > 1$.
 - is associative
- composition on $\{f: A \rightarrow A\}$:
 - not commutative
 - is associative

$$(f \circ g) \circ h \stackrel{?}{=} f \circ (g \circ h)$$

$$(f \circ g)(h(x)) \stackrel{?}{=} f((g \circ h)(x))$$

$$\longrightarrow f(g(h(x)))$$

Tables of binary operations

Determine a binary operation $*$ on a finite set S by writing all values $a * b$ in a table.

Example: $S = \{a, b, c\}$

$*$	a	b	c
a	b	c	b
b	a	c	b
c	c	b	a

$$a * b = c$$

$$c * c = a$$

Q: why does a table define a binary operation?

Testing commutativity

Prop: $*$ is commutative if and only if its table is symmetric about the main diagonal

$*$	a	b	c
a	b	c	b
b	a	c	b
c	c	b	a

Not commutative

Isomorphic Binary Structures

Section 3

When are two binary operations structurally the same?

Def: We call $(S, *)$ a **binary algebraic structure** if $*$ is a **binary operation** on S .

Def: An **isomorphism** of $(S, *)$ with $(S', *')$ is a **one-to-one** and onto function $\phi: S \rightarrow S'$ such that:

surjective $\uparrow$

$$\forall x, y \in S: \phi(x * y) = \phi(x) *' \phi(y)$$

homomorphism property $\nwarrow$

we say $(S, *)$ and $(S', *')$ are **isomorphic** and write

$$(S, *) \cong (S', *') \text{ OR } S \cong S'$$

a one-to-one and onto function is called a **bijection**

Steps to show $(S, *) \simeq (S', *')$:

① Define a function $\phi: S \rightarrow S'$.

② Show ϕ is injective: $\phi(x) = \phi(y) \Rightarrow x = y$

③ Show ϕ is surjective: $\forall x' \in S', \exists x \in S$ with $\phi(x) = x'$.

④ Show ϕ satisfies the homomorphism property:

$$\forall x, y \in S, \quad \phi(x * y) = \phi(x) *' \phi(y).$$

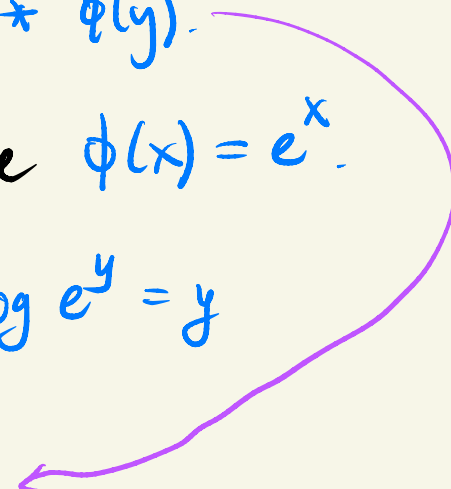
Example: $(\mathbb{R}, +) \simeq (\mathbb{R}^+, \cdot)$. Define $\phi(x) = e^x$.

① ϕ is well defined and lands in $\mathbb{R}^+$

② $\nexists e^x = e^y$ then $x = \log e^x = \log e^y = y$

③ let $x \in \mathbb{R}^+$, then $e^{\log x} = x$.

④ $e^{x+y} \stackrel{?}{=} e^x \cdot e^y$



How to show $(S, *)$ and $(S', *')$ are NOT isomorphic

Find a **structural property** on which they differ:

a property that $(S, *)$ and $(S', *')$ would share if they were isomorphic.

Examples: of structural properties

- cardinality of S
- $*$ commutative or associative
- $\vdots$
- existence of an identity element

Non-Examples:

- -1 is an element
- operation is $+$
- elements are matrices

Identity elements

Def: Let $(S, *)$ be a binary algebraic structure. An element $e \in S$ is called an **identity element** if:

$$e * x = x * e = x, \quad \forall x \in S.$$

Thm: If $(S, *)$ has an identity element, it is unique.

Pf: $\nexists e_1, e_2$ are both identity elements

$$e_1 = e_1 * e_2 = e_2$$

$\uparrow$
 e_2 is
identity

$\uparrow$
 e_1 is identity



Examples:

• $(\mathbb{Z}, +)$: $e = 0$, since $0 + x = x + 0 = x, \forall x \in \mathbb{Z}$

• $(\mathbb{R}, \cdot)$: $e = 1$, $1 \cdot x = x \cdot 1 = x, \forall x \in \mathbb{R}$

• $\text{Mat}_n(\mathbb{C})$: $e = \begin{pmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{pmatrix}$ identity matrix
under $\times$

Thm: If $\phi: S \rightarrow S'$ is an isomorphism and S has an identity e , then $\phi(e)$ is an identity for S' .

Pf: let $x' \in S'$.

find x s.t.
 $\phi(x) = x'$
↓

$$\phi(e) *' x' = \phi(e) *' \phi(x) = \phi(e * x) = \phi(x) = x' \quad \square$$

Admin

- Lecture notes will be posted (but not recordings)
 - Please come to class!
- HW 1 posted, due 9/1 (after that ~weekly)
 - due by 2 PM
 - HW is best way to engage with class material

AI Use it to study, not think for you

- Quiz 1 on 9/2 in Discussion
 - 4 questions, multiple choice

Read the syllabus